UNIVERSITY OF UYO, UYO
FACULTY OF PHYSICAL SCIENCES
DEPARTMENT OF MATHEMATICS
2022/2023 FIRST SEMESTER EXAMINATION
MTH 316: INTRODUCTION TO MATHEMATICAL MODELLING

Instruction: Answer **Question 1** and any other **TWO** questions **Time: 2hrs**

1(a) What is mathematical modelling?
(b) Explain briefly why mathematical models are relevant in sciences?
(c) Discuss three fundamental assumptions upon which mathematical models are based.
(d) A population of antelopes in a limited environment is given by a continuous logistic model

$$A' = kA(\varphi - A)$$

where $A(t)$ is the number of antelopes at any time t, k is a positive constant and φ is the carrying capacity of the environment.

(i) Obtain an expression for the population of antelopes $A(t)$ at any time t
(ii) Demonstrate that as time passes, the population of antelopes will eventually equal φ

2 A CSI agent arrives 15 minutes after a crime of murder has been reported. He then proceeds to some preliminary scene inspection for another 10 minutes before resuming work on the body of the victim found. The room and body temperature were noted to be 75°F and 90°F respectively at exactly 8:45pm. The body temperature dropped by 10°F as the agent was leaving the scene at 9:45pm.

(a) State clearly some assumptions that must be made to simplify the mathematical model that captures the described scenario.
(b) Formulate the model and use it to estimate the time of death. [Hint: Normal body temperature is given as 98.6°F]

3 The reproduction of a certain bacteria is given by the discrete model; $x_{n+1} = mx_n$ where m is itself a function of x_n, given as; $m = \frac{2}{1+{}^{x_n}/_{k}}$

(a) Show that the solution of the model is $x_n = \frac{1}{\frac{1}{2^n x_0} + \frac{1-(1/2)^n}{k}}$
(b) What happens as n approaches infinity?

4(a) Write the stoichiometric matrix associated with the chemical reactions and hence obtain the corresponding system of differential equations modelling the chemical kinetics below.

$$A \rightleftharpoons B \rightleftharpoons C$$

where the reaction A to B is k_1, B to C is k_2, B to A is k_3 and C to B is k_4.

(b) A tank initially contains 120L of pure water. A mixture containing a concentration of γ g/L of salt enters the tank at a rate of 2 L/min, and the well-stirred mixture leaves the tank at the same rate.
(i) Obtain an expression in terms of γ for the amount of salt in the tank at any time t.
(ii) Find the limiting amount of salt in the tank as $t \to \infty$.

5 Suppose certain pollutants are emptied into your local river and $P(x,t)$ denotes the concentration level of pollutants discharged some x miles away at any time t.

(a) Formulate (without solving) the model for dispersion of pollution in the river.
(b) Set up a suitable initial and boundary conditions for your model.

$A \to B = k_1$
$B \to C = k_2$
$B \to A = k_3$
$C \to B = k_4$

$\therefore t = k_1 + k_2 - k_4$